A NEGATIVE ANSWER TO A QUESTION ON MONOTONE MAXIMAL CHAINS IN FINITE GROUPS

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ABSTRACT. In correspondence beginning on August 8, 2026, V. S. Monakhov and I. L. Sokhor communicated the following question to the author: does every finite group have an unrefinable chain of subgroups whose successive indices are nondecreasing? We answer this question in the negative, even among soluble groups. For every odd prime p, we construct a soluble matrix group $G_p \leq \mathrm{GL}_5(p)$ of order 2^6p^8 with no such chain. These counterexamples form an infinite family whose smallest member, G_3, has order 419904. The construction is based on an example due to Kohler, and the obstruction follows from three elementary calculations of maximal-subgroup indices.

1. Introduction

Let G be a nontrivial finite group. An unrefinable chain of subgroups

$$1 = G_0 < G_1 < \cdots < G_n = G$$

is one in which G_{i-1} is maximal in G_i for every i. Associate with this chain the indices $j_i = |G_i : G_{i-1}|$. We call the chain increasing if

$$j_1 \leq j_2 \leq \cdots \leq j_n.$$

Monakhov and Sokhor call such a chain a $(<)$-chain [2]. In correspondence beginning on August 8, 2026, V. S. Monakhov and I. L. Sokhor communicated to the author the following question, stated here in equivalent terminology.[1]

> Does every finite group contain an increasing unrefinable subgroup chain?

We answer this question in the negative.

Theorem 1. *For every odd prime p, there is a soluble group G_p of order 2^6p^8 which has no increasing unrefinable subgroup chain.*

The smallest group in the family has order $2^63^8 = 419904$. We do not claim that it is the smallest counterexample. Our construction is based on the $t = 2$ case of Kohler's work on maximal-subgroup indices [1]. Kohler used this construction to show that bounds on the indices of maximal subgroups of a finite group need not be inherited by its subgroups. In the matrix model below, the four coordinates occurring in x and y correspond to four degree-one generators, while the four entries of z record the independent cross-commutators. The resulting p-group therefore has order $p^{4+4} = p^8$. We exploit the resulting nested pattern of maximal-subgroup indices.

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[1] The correspondence is the source through which the author learned of the question; no claim is made here about the problem's original formulation. Reference [2] is cited for terminology and background, not as a published source of the universal-existence question.

2. A lemma on maximal subgroups

For a finite group A, write

$$\mathcal{I}(A) = \{|A : B| \mid B \text{ is maximal in } A\}.$$

Lemma 2. *Let V be an elementary abelian p-group, let H be a p'-group acting on V, put $K = V \rtimes H$, and let $\pi : K \to H$ be the natural projection. Every maximal subgroup of K has one of the following forms:*

(1) $\pi^{-1}(J) = V \rtimes J$, *where J is maximal in H;*
(2) *a V-conjugate of $W \rtimes H$, where W is a maximal proper H-submodule of V.*

Proof. Let T be maximal in K. If $V \leq T$, then T/V is maximal in $K/V \cong H$, which gives the first form. Suppose $V \not\leq T$. Then $VT = K$. Put $W = V \cap T$. The group T normalizes W, while V centralizes W; hence $W \lhd K$. In K/W, both T/W and WH/W are complements to V/W. By the Schur–Zassenhaus theorem, these complements are conjugate by an element of V/W. Lifting this conjugacy shows that T is V-conjugate to $W \rtimes H$. Moreover, maximality of T is equivalent to maximality of W among the proper H-submodules of V. $\square$

In particular, if H is a 2-group, the subgroups of the first type have index 2. If V is completely reducible, the indices of the subgroups of the second type are the orders of the irreducible quotients of V.

3. The counterexample family

Fix an odd prime p and put $F = \mathbb{F}_p$. Let

$$s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad D = \langle s, t \rangle \leq \mathrm{GL}_2(F).$$

Then $D \cong D_8$, where D_8 has order 8. The natural FD-module $U = F^2$ is absolutely irreducible. Indeed, after extending scalars to an algebraic closure, the only t-invariant lines are the two coordinate lines, and s interchanges them.

Let $H = D \times D$. Define

$$L = \left\{ \ell(x,y,z) = \begin{pmatrix} I_2 & x & z \\ 0 & 1 & y \\ 0 & 0 & I_2 \end{pmatrix} : x \in F^{2\times 1},\ y \in F^{1\times 2},\ z \in F^{2\times 2} \right\}.$$

Here the block sizes are $2, 1, 2$. Direct multiplication yields

$$\ell(x,y,z)\ell(x',y',z') = \ell(x+x', y+y', z+z'+xy'). \tag{1}$$

With the convention $[a,b] = a^{-1}b^{-1}ab$, it follows that

$$[\ell(x,y,z), \ell(x',y',z')] = \ell(0,0,xy'-x'y). \tag{2}$$

Thus $|L| = p^8$. Put

$$Z = \{\ell(0,0,z)\}, \quad X = \{\ell(x,0,0)\}, \quad Y = \{\ell(0,y,0)\}.$$

The group Z is central in L. The outer products xy span $F^{2\times 2}$, so (2) gives $L' = Z$. Since p is odd, L has exponent p; explicitly,

$$\ell(x,y,z)^p = \ell\left(0,0,pz + \binom{p}{2} xy\right) = 1.$$

The standard formula for the Frattini subgroup of a finite p-group now gives

$$\Phi(L) = L^p L' = Z. \tag{3}$$

Embed H in $\mathrm{GL}_5(F)$ by identifying $(A,B)\in D\times D$ with the block-diagonal matrix $\operatorname{diag}(A,1,B)$. Conjugation by this matrix acts by

$$x\mapsto Ax,\qquad y\mapsto yB^{-1},\qquad z\mapsto AzB^{-1}.$$

Consequently, H normalizes both L and Z. Define

$$G_p = L\rtimes H \le \mathrm{GL}_5(p).$$

Then $Z\lhd G_p$ and $|G_p| = p^8\cdot 8^2 = 2^6p^8$. Since both L and H are soluble, so is G_p.

Set $\overline{X} = XZ/Z$ and $\overline{Y} = YZ/Z$. As an H-module,

$$L/Z = \overline{X}\oplus\overline{Y}, \tag{4}$$

where $\overline{X}$ and $\overline{Y}$ are nonisomorphic irreducible modules of dimension 2. More explicitly, the first factor of $D\times D$ acts naturally on $\overline{X}$, while the second acts trivially. The second factor acts dually on $\overline{Y}$, while the first acts trivially. Thus

$$C_H(\overline{X}) = 1\times D,\qquad C_H(\overline{Y}) = D\times 1.$$

These action kernels also show that the two modules are nonisomorphic. Since $p\nmid |H| = 64$, Maschke's theorem applies. Hence the only maximal H-submodules of L/Z are $\overline{X}$ and $\overline{Y}$.

Moreover,

$$Z\cong U\boxtimes U^* \tag{5}$$

is the external tensor product for the two direct factors of H. Hence Z is an irreducible H-module of dimension 4. Indeed, over an algebraic closure $\overline{F}$, both $U\otimes_F\overline{F}$ and $U^*\otimes_F\overline{F}$ remain irreducible, and their external tensor product is irreducible for $D\times D$.

4. The index obstruction

Lemma 3. *The maximal-index spectra are*

$$\begin{aligned}\mathcal{I}(G_p) &= \{2,p^2\},\\ \mathcal{I}(M_X) = \mathcal{I}(M_Y) &= \{2,p^2,p^4\},\\ \mathcal{I}(N) &= \{2,p^4\},\end{aligned}$$

where

$$M_X = (Z\times X)\rtimes H,\qquad M_Y = (Z\times Y)\rtimes H,\qquad N = Z\rtimes H.$$

Every maximal subgroup of G_p of index p^2 is G_p-conjugate to M_X or M_Y. Every maximal subgroup of M_X of index p^2 is M_X-conjugate to N, and every maximal subgroup of M_Y of index p^2 is M_Y-conjugate to N.

Proof. First, we show that every maximal subgroup M of G_p contains Z. If $Z\not\le M$, then maximality gives $G_p = ZM$, and the modular law together with (3) gives

$$L = L\cap ZM = Z(L\cap M) = \Phi(L)(L\cap M).$$

Recall the standard Frattini fact that $L=\Phi(L)K$ for a subgroup $K\le L$ implies $K=L$: otherwise, a maximal subgroup of L containing K would contain both K and $\Phi(L)$, and hence all of L, a contradiction. Applying this fact with $K = L\cap M$ forces $L\cap M = L$, so $Z\le M$, again a contradiction.

It follows that the maximal subgroups of G_p are the inverse images of the maximal subgroups of

$$G_p/Z = (\overline{X}\oplus\overline{Y})\rtimes H.$$

Apply Lemma 2 and (4). The maximal subgroups of the first type have index 2, while those of the second type have index p^2. This proves the first equality and the assertion concerning M_X and M_Y.

In M_X, the normal elementary abelian subgroup $Z \times X$ is the direct sum of nonisomorphic irreducible H-modules of dimensions 4 and 2. Its only maximal H-submodules are therefore Z and X. Another application of Lemma 2 gives the indices $2, p^2, p^4$; the index-p^2 case retains Z and gives an M_X-conjugate of N. The same argument applies to M_Y. Finally, Z is irreducible of dimension 4, so Lemma 2 applied to $N = Z \rtimes H$ gives $\mathcal{I}(N) = \{2, p^4\}$. □

Proof of Theorem 1. Suppose

$$1 = G_0 < G_1 < \cdots < G_n = G_p$$

is unrefinable and its indices $j_i = |G_i : G_{i-1}|$ are nondecreasing. By Lemma 3, $j_n \in \{2, p^2\}$. We cannot have $j_n = 2$: then monotonicity would force every j_i to be 2, whereas $p \mid |G_p|$. Thus $j_n = p^2$, and G_{n-1} is G_p-conjugate to M_X or M_Y.

Now $j_{n-1} \le p^2$. The middle spectrum in Lemma 3 shows that j_{n-1} is 2 or p^2. It cannot be 2, by the same divisibility argument applied to G_{n-1}. Hence $j_{n-1} = p^2$, and G_{n-2} is G_p-conjugate to N.

Finally, $j_{n-2} \le p^2$. Since G_{n-2} is G_p-conjugate to N and $\mathcal{I}(N) = \{2, p^4\}$, we must have $j_{n-2} = 2$. This forces all preceding indices to equal 2, which is impossible because $p \mid |N|$. The contradiction proves the theorem. □

5. The fully explicit group G_3

To facilitate independent verification, we give the smallest member of the family without parameters. All matrices below are over $\mathbb{F}_3$, so $-1 = 2$. Let E_{ij} denote the standard matrix unit, write $e_{ij} = I_5 + E_{ij}$, and set

$$a_s = \mathrm{diag}(s, 1, I_2), \quad a_t = \mathrm{diag}(t, 1, I_2),$$
$$b_s = \mathrm{diag}(I_2, 1, s), \quad b_t = \mathrm{diag}(I_2, 1, t).$$

Then the concrete example is the following subgroup of $\mathrm{GL}_5(3)$:

$$G_3 = \langle e_{13}, e_{23}, e_{34}, e_{35}, a_s, a_t, b_s, b_t \rangle. \tag{6}$$

Define

$$\begin{aligned} Z_3 &= \langle e_{14}, e_{15}, e_{24}, e_{25} \rangle, \\ H_3 &= \langle a_s, a_t, b_s, b_t \rangle, \\ M_{X,3} &= \langle Z_3, e_{13}, e_{23}, H_3 \rangle, \\ M_{Y,3} &= \langle Z_3, e_{34}, e_{35}, H_3 \rangle, \qquad N_3 = \langle Z_3, H_3 \rangle. \end{aligned}$$

Here $|H_3| = 64$. The orders and complete maximal-index spectra relevant to the obstruction are

A	$\lvert A\rvert$	$\mathcal{I}(A)$
G_3	419904	$\{2, 9\}$
$M_{X,3}, M_{Y,3}$	46656	$\{2, 9, 81\}$
N_3	5184	$\{2, 81\}$

The standalone file `G3-verification.g` accompanying this manuscript defines all eight generators in (6) as explicit 5×5 matrices. Using only standard GAP commands, it verifies the group and subgroup orders, the five named maximal inclusions, every conjugacy class of maximal subgroups and its index, the coverage of all relevant classes by the named subgroups, and the result of an exhaustive top-down search for an increasing chain. The file has been run unchanged under GAP 4.11.1 and GAP 4.16.0. Thus the computation can be checked from the attached file without access to an online repository.

Remark 4. *GAP computations for $p = 3, 5, 7$ confirm the predicted spectra; for $p = 3$ these are*

$$\{2, 9\}, \qquad \{2, 9, 81\}, \qquad \{2, 81\},$$

and an exhaustive branch search finds no increasing unrefinable chain. These computations corroborate the theorem but are not used in its proof. The standalone $p = 3$ file, complete code, tests, and captured outputs are publicly available in release v1.1.0 of the accompanying repository [3].

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Code and data availability

The complete computational supplement is publicly available in the verification repository. Release `v1.1.0` is the version corresponding to this manuscript. It includes the standalone verifier and captured outputs from GAP 4.11.1 and GAP 4.16.0. Source code is licensed under the MIT License, and the manuscript and documentation are licensed under CC BY 4.0.

Declaration of generative AI and AI-assisted technologies

During this project the author used Anthropic's Claude and OpenAI Codex for mathematical exploration and criticism, literature-search assistance, drafting and debugging GAP code, and manuscript organization and editing. The author independently checked the mathematical argument, reran all tests in the repository, and verified the cited references. He takes full responsibility for the content of the article.

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